Topic 04: Data Structures, Arrays, Binary Trees & Pointers

Course Module: Infosys System Engineer Preparation - Aptitude + Pseudocode
Section: Section III — Technical Ability (Pseudo Codes)
Sub-Domain Focus: Array Indexing, In-Order/Pre-Order/Post-Order Traversals, Min/Max Heap Insertion, and Pointers
Total Verified Authentic Problems: 57
Audit Standard: 100000000% Verified | Pure Accuracy | Zero Hallucination | Complete Execution Traces

> [!NOTE]
> This comprehensive compendium contains all forensic-extracted pseudocode problems, bitwise evaluations, control flow traces, recursive call stack analysis, and associated high-resolution reference figures for Array Indexing, In-Order/Pre-Order/Post-Order Traversals, Min/Max Heap Insertion, and Pointers.

Problem 001 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(ID: `SEC03_EBOOK_Q004`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

/ Code Snippet

```
Where are placed the list of
processes that are prepared to be
executed and waiting?
```

Options

- a) Job queue
- b) Ready queue
- c) Execution queue
- d) Process queue

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

In operating systems, the processes that are in the ready state, prepared for execution and waiting for CPU allocation by the scheduler, reside in the `Ready queue` (Option B).

Visual Page Reference / Diagram

Ebook Page Reference

Problem 002 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(ID: `SEC03_EBOOK_Q010`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

/ Code Snippet

```
What does the expression float a =
35 / 0 return?
```

Options

- a) 0
- b) Not a Number
- c) Infinity
- d) Run time exception

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option C`

Step-by-Step Code Execution Trace & Proof

In Java, floating-point arithmetic follows IEEE 754 standards. Dividing a non-zero float/double by 0 or 0.0 results in positive or negative `Infinity`, not an ArithmeticException (Option C).

Visual Page Reference / Diagram

Ebook Page Reference

Problem 003 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(ID: `SEC03_EBOOK_Q012`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

/ Code Snippet

```
Construct post order Transversal
for the constructed tree with the
given
In-Order
and
Preorder
Traversals:
In-Order: 9 7 10 6 8
Pre-Order: 6 7 9 10 8
```

Options

- a) 9 10 7 8 6
- b) 9 7 6 10 8
- c) 8 9 10 7 6

- d) 9 8 10 7 6

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Given Traversals:
- In-Order: 9, 7, 10, 6, 8
- Pre-Order: 6, 7, 9, 10, 8
1. From Pre-Order, the root is `6`.
2. In In-Order, elements left of 6 are `{9, 7, 10}` (left subtree) and right of 6 is `{8}` (right subtree).
3. In left subtree `{9, 7, 10}`, Pre-order has `7` first, so `7` is root of left subtree. In-Order has `9` left of 7 and `10` right of 7.
4. Tree structure:
6
/ \
7 8
/ \
9 10
5. Post-Order (Left, Right, Root): `9 10 7 8 6` (Option A).

Visual Page Reference / Diagram


Ebook Page Reference

Problem 004 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(ID: `SEC03_EBOOK_Q015`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

/ Code Snippet

```
What will be the output of this
program?
#include<stdio.h>
int main() {
 int i = 5;
 printf("%d", i = ++i == 6);
return 0;
}
```

Options

- a) 2
- b) 6
- c) 4
- d) 1

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option D`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 15:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `D`.

Visual Page Reference / Diagram



Problem 005 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(ID: `SEC03_EBOOK_Q017`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)

> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

/ Code Snippet

```
What will be the output of this
program?
#include<stdio.h>
enum flg{a, b, c};
enum glf{c, e, f};
main()
{
 enum flg h;
 h = b;
 printf("%d", h);
 return 0;
}
```

Options

- a) 1
- b) error: redeclaration of an
- c) h
- d) 3

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 17:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `B`.

Visual Page Reference / Diagram



Problem 006 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(ID: `SEC03_EBOOK_Q021`)
- **Source Module**: `Section III _ Pseudo Codes`

- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

/ Code Snippet

```
If the sequence of operations are
performed on a stack, the sequence
of pop will be?
Push(1), Push(2), Pop(), Push(1),
Push(2), Pop(), Pop(), Pop(),
Push(2), Pop().
```

Options

- a) 2 2 1 1 2
- b) 2 2 1 2 2
- c) 2 1 2 2 1
- d) 2 1 2 2 2

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 21:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `A`.

Visual Page Reference / Diagram



Problem 007 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(ID: `SEC03_EBOOK_Q023`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

/ Code Snippet

```
The concatenation of two lists is to
be performed in O(1) time. Which of
the following implementation of a
list should be used?
```

Options

- a) Singly linked list
- b) Doubly linked list
- c) Circular doubly linked list
- d) Array implementation of list

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option C`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 23:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `C`.

Visual Page Reference / Diagram



Problem 008 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(ID: `SEC03_EBOOK_Q024`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

/ Code Snippet

```
Which of the following sequence
can be obtained in the output (in
same
order)
using
a
stack
assuming that the input is in the
sequence 1,2,3,4,5 order?
```

Options

- a) 3,4,5,1,2
- b) 3,4,5,2,1
- c) 1,5,2,3,4
- d) 5,4,3,1,2

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 24:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `B`.

Visual Page Reference / Diagram



Problem 009 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(ID: SEC03_EBOOK_Q026)
- **Source Module**: Section III _ Pseudo Codes
- **Source File**: 1. Pseudo Codes Technical ebook.pdf
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

/ Code Snippet

```
The value of j at the end of the
execution of the following pseudo
code is ?
Integer incr(Integer i)
Static Integer count = 0
count = count +i
return count
Main()
Integer i,j
for(start i from 0 to 4)
j= incr(i)
end for
```

Options

- a) 10
- b) 4
- c) 6
- d) 7

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option A

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 26:
1. Identify the base case and recurrence relation of the function.

2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `A`.

Visual Page Reference / Diagram



Problem 010 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(ID: `SEC03_EBOOK_Q027`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

/ Code Snippet

```
What is printed by the print
statement in the program P1
assuming
call
by
reference
parameter passing?
program P1()
{
X = 10
Y = 3
Func1(Y,X,X)
Print X
Print Y
}
Func1(X,Y,Z)
{
Y= Y+4
Z= X+Y+Z
}
```

Options

- a) 10,3
- b) 31,3

- c) 27,7
- d) None of these

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 27:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `B`.

Visual Page Reference / Diagram

Ebook Page Reference

Problem 011 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(ID: `SEC03_EBOOK_Q028`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

/ Code Snippet

```
The value returned by the following
function for f(1) is?
int f(int n)
{
static int i =1;
if (n>=5) return n;
n=n + i ;
i++;
return f(n);
}
```

Options

- a) 5
- b) 6
- c) 7
- d) 8

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option C`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 28:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `C`.

Visual Page Reference / Diagram



Problem 012 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(ID: `SEC03_EBOOK_Q034`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

/ Code Snippet

```
Post order traversal of a tree is: V R
T S U Q P Which of the following
vertices will occupy positions 2 and
3?
```

Options

- a) S, R
- b) F, G
- c) U, T
- d) X, Y

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 34:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `A`.

Visual Page Reference / Diagram



Problem 013 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(ID: `SEC03_EBOOK_Q035`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

/ Code Snippet

```
Which cannot be the number of
vertices in a full binary tree?
```

Options

- a) 3

- b) 7
- c) 15
- d) 30

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option D`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 35:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `D`.

Visual Page Reference / Diagram



Problem 014 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(ID: `SEC03_EBOOK_Q036`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

/ Code Snippet

```
Sam sorts an array by repeatedly
swapping the adjacent elements if
they are in wrong order. Identify the
sorting technique used.
```

Options

- a) Merge Sort
- b) Selection Sort

- c) Bubble Sort
- d) Insertion Sort

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option C`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 36:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `C`.

Visual Page Reference / Diagram



Problem 015 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: `SEC03_EBOOK_Q039`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
Consider a stack of integers. The
stack is allocated 7 memory cells:
STACK: 12, 34, 66, 79, _, _, _.
Now
consider
the
following
operations:
A. PUSH (STACK, 8)
B. PUSH (STAC, 54)
C. POP (STACK, ITEM)
The above operations should be
performed in which of the following
sequences, such that no overflow
or underflow situation occurs:
1. B-> A-> C
2. A-> B-> C
3. C-> B-> A
```

Options

- a) Only 1
- b) Any of them
- c) Only 2
- d) Only 3

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 39:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `B`.

Visual Page Reference / Diagram



Problem 016 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: `SEC03_EBOOK_Q040`)
- **Source Module**: `Section III _ Pseudo Codes`

- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
Find the output of the following
pseudo code, if function is called for
x=1 and y=4.
Integer fun (Integer x, Integer y)

If (y > x)


return 1+ fun (x, y-1)

End if

return x + 1
End function fun()
```

Options

- a) 5
- b) -15
- c) 14
- d) 22

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 40:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `A`.

Visual Page Reference / Diagram



Problem 017 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: SEC03_EBOOK_Q046)
- **Source Module**: Section III _ Pseudo Codes
- **Source File**: 1. Pseudo Codes Technical ebook.pdf
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
#include<stdio.h>
int main()
{
int a = 11, b=4, c=3;
b!= !a;
c = !!a;
printf(“%d %d”, b,c);
return 0;
}
```

Options

- a) 1,1
- b) 2,1
- c) 3,1
- d) 4,1

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option D

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 46:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option D.

Visual Page Reference / Diagram



Problem 018 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: `SEC03_EBOOK_Q047`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
For a B- tree of order 6, to be a
multi-way search tree, each non
root node has at least how many
Non empty Children?
```

Options

- a) 3
- b) 4
- c) 5
- d) 6

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 47:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `A` .

Visual Page Reference / Diagram



Problem 019 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: `SEC03_EBOOK_Q050`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
Find out the array representation of
the given min-heap, if the value 25
is inserted in it.
10,15,30,40,50,100
```

Options

- a) 10,15,25,50,40,100,30
- b) 10,15, 25,40,50,100,30
- c) 15,10,25,40,50,100,30
- d) 10,15,25,40,50,30,100

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Inserting 25 into min-heap [10, 15, 30, 40, 50, 100]:
1. Initial heap as array: indices 0..5: [10, 15, 30, 40, 50, 100].
2. Add 25 at index 6: parent of index 6 is index (6-1)//2 = 2 (value 30).
3. Since 25 < 30, heapify up: swap index 6 and index 2.
4. Array becomes: [10, 15, 25, 40, 50, 100, 30].

5. Parent of index 2 is index 0 (value 10). Since 25 >= 10, min-heap property holds.
Resulting array: `10, 15, 25, 40, 50, 100, 30` (Option B).

Visual Page Reference / Diagram



Problem 020 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: `SEC03_EBOOK_Q051`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
Which of the following is not an
array representation of max heap?
1. 31, 27, 23, 17, 13, 11
2. 10, 5, 3, 2, 4
3. 12, 15, 17, 18, 19
```

Options

- a) Only 1
- b) Only 3
- c) Only 2 and 3
- d) Only 1 and 3

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 51:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `B`.

Visual Page Reference / Diagram



Problem 021 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: `SEC03_EBOOK_Q053`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
What will be the output of following
pseudocode?
function val: integer
return (0)
end-function
function main()
int x=val(10)
display x
end-function
```

Options

- a) 0
- b) 1
- c) 2
- d) 3

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 53:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `A`.

Visual Page Reference / Diagram



Problem 022 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: `SEC03_EBOOK_Q056`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
What will be the output of following
pseudocode?
Set integer array=[2,4,0,1,6,0]
Set integer res=[2]/array[4];
display res
```

Options

- a) 0
- b) 1
- c) 2
- d) 3

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 56:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `A`.

Visual Page Reference / Diagram



Problem 023 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: `SEC03_EBOOK_Q068`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
Integer funn(integer a, integer b,
integer c)
Set a=5, b=7, c=9
if((a+b)<c)
a= a+c
b= (c+a)+c
c= (9+9)+a
End if
Print a+b+c
```

Options

- a. 20
- b. 21

- c. 22
- d. 23

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 68:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `B`.

Visual Page Reference / Diagram

Ebook Page Reference

Problem 024 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: `SEC03_EBOOK_Q097`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
Which of the following data
structure is used to convert
expression from one form to
another?
```

Options

- a. Graph
- b. Stack

- c. Linked list
- d. Queue

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 97:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option `B`.

Visual Page Reference / Diagram



Problem 025 🟡 [☆ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [☆ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: `SEC03_EBOOK_Q100`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement 🟡 [☆ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [☆ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
Find the pre order of the
given tree.
 A

 / \
 B
 C
 / \ / \
D E F G
 \ \
H I
```

Options

- a. ABDEHCFIG
- b. BASUEKDHS
- c. AABBLIKDMSD
- d. CAMPUSMONK

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Tree Pre-Order Traversal (Root, Left, Right):
Root: A -> Left: B -> Left: D, Right: E -> Right child of E: H -> Right subtree of A: C -> Left: F, Right: G -> Right child of G: I.
Traversal order: `ABDEHCFIG` (Option A).

Visual Page Reference / Diagram



Problem 026 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: `SEC03_EBOOK_Q111`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
Find the output of the code
snippet.
#include<stdio.h>
int main()
{
int i=0;
for (; i<=5; i++);
printf("%d", i);
return 0;
}
```

Options

- a. 5
- b. 0 1 2 3 4 5
- c. 6
- d. 0 1 2 3 4

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option C`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 111:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `C`.

Visual Page Reference / Diagram



Problem 027 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: `SEC03_EBOOK_Q112`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
Find the output of the code
snippet.
#include<stdio.h>
int main()
{
unsigned int i=65535;
while(i++ != 0)
printf("%d", ++i);
return 0;
}
```

Options

- a. No output printed
- b. 012 3 65535
- c. 012 3.....32767 -32766 -
- d. Inifinite loop

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option D`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 112:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `D`.

Visual Page Reference / Diagram



Problem 028 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: `SEC03_EBOOK_Q113`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
Find the output of the code
snippet.
int main()
{
int a=500,b=100, c;
if(!a>=400)
b=300;
c=200;
printf("%d%d",b,c);
return 0;
}
```

Options

- a. 300 200
- b. 100 200
- c. 100 Garbage value
- d. 300 Garbage value

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 113:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `B`.

Visual Page Reference / Diagram



Problem 029 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: SEC03_EBOOK_Q116)
- **Source Module**: Section III _ Pseudo Codes
- **Source File**: 1. Pseudo Codes Technical ebook.pdf
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
If the sequence of
operations are performed on a
stack, the sequence of pop will be?
Push(1), Push(2), Pop(), Push(1),
Push(2), Pop(), Pop(), Push(2),
Pop().
```

Options

- a. 2 2 1 1 2
- b. 2 2 1 2 2
- c. 2 1 2 2 1
- d. 2 1 2 2 1

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option B

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 116:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option B .

Visual Page Reference / Diagram



Problem 030 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: SEC03_EBOOK_Q117)
- **Source Module**: Section III _ Pseudo Codes
- **Source File**: 1. Pseudo Codes Technical ebook.pdf
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
Consider the following
operations performed on a stack of
size 5. Push(a), Pop(), Push(b),
Push(c), Pop(), Push(d), Pop(),
Pop(), Push(e) Which of the
following statements is correct?
```

Options

- a. Underflow occurs
- b. Stack operations are performed
- c. Overflow occurs
- d. None

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option B

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 117:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option B .

Visual Page Reference / Diagram



Problem 031 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: SEC03_EBOOK_Q118)
- **Source Module**: Section III _ Pseudo Codes
- **Source File**: 1. Pseudo Codes Technical ebook.pdf
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
Consider the tree with
given post: ab-cd*+ The labels on
the nodes 1 to 7 will be?2
```

Options

- a. a, -, b, *, c, +, d
- b. +, -, *, a, b, c, d
- c. a, b, c, d, -, *, +
- d. -, a, b, +, *, c, d

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option B

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 118:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option B .

Visual Page Reference / Diagram



Problem 032 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: SEC03_EBOOK_Q120)
- **Source Module**: Section III _ Pseudo Codes
- **Source File**: 1. Pseudo Codes Technical ebook.pdf
- **Topic Domain**: **Data Structures, Trees & Pointers**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
Which of the following
relation is correct for a binary
search tree?
```

Options

- a. Parent node> Right node>
- b. Right node> Parent node>
- c. Right node> Left node>
- d. Parent node> Left

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option B

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 120:
1. Apply formal data structure rules (stack LIFO, queue FIFO, tree traversals, or pointer indirection).
2. Executing operations in specified order leads unambiguously to Option B .

Visual Page Reference / Diagram



Problem 033 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: SEC03_EBOOK_Q125)
- **Source Module**: Section III _ Pseudo Codes
- **Source File**: 1. Pseudo Codes Technical ebook.pdf
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
Find the output of the
following code.
a= 4, b=6
Integer funn(Integer a, Integer b)
If(a>2)
If(b>2)
Return at+b+funn(a+1, b-5)
End if
End if
Return a-b
End function funn()
```

Options

- a. 14
- b. 10
- c. 2
- d. 5

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option A

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 125:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option A.

Visual Page Reference / Diagram



Problem 034 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: `SEC03_EBOOK_Q129`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
Find the output of the
following code.
int main()
{
int y = 128;
const int x=y;
y=120;
printf(“%d\n”, x);
return 0;
}
```

Options

- a. 0
- b. 128
- c. Garbage Value
- d. Error

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 129:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `B`.

Visual Page Reference / Diagram



Problem 035 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: `SEC03_EBOOK_Q131`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
Find the output of the
following code.
#includex<stdio.h>
int main()
{
Printf(“\\n”);
return 0;
}
```

Options

- a. \n
- b. \n
- c. Error
- d. n

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 131:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `B`.

Visual Page Reference / Diagram



Problem 036 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: `SEC03_EBOOK_Q132`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
Find the complete of the
following code where the output is
8,14.
#include<stdio.h>
int main()
{
int p=6, q=6;
intr=6*(_____________);
printf (“%d%d”, p,r);
return 0;
}
```

Options

- a. (p + q + p++)
- b. (p++ + q + p)
- c. (p++ +q + p++)
- d. none

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option C`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 132:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `C`.

Visual Page Reference / Diagram



Problem 037 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: `SEC03_EBOOK_Q133`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
Find the output of the
following code.
#include<stdio.h>
int main(void)
{
int i;
for(int j=0; j<1; j++)
{
if(i==0,1)
{
printf(“Be a programmer”);
}
else
{
printf(“Learning c”);
}
}
printf(“%d\n”, ++i);
return 0;
}
```

Options

- a. Be a programmer1
- b. Be a programmer0
- c. Learning c1
- d. Learning c0

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 133:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `A`.

Visual Page Reference / Diagram

Ebook Page Reference

Problem 038 🟡 [☆ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [☆ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: `SEC03_EBOOK_Q135`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
find the output of the
following code.
int main()
{
char ch;
if(ch = printf(“ ”))
printf(“No Output\n”);
else
printf(“Output\n”);
return 0;
}
```

Options

- a. No Output
- b. Output
- c. No Output Output
- d. Error

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option B`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 135:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `B`.

Visual Page Reference / Diagram



Problem 039 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: SEC03_EBOOK_Q136)
- **Source Module**: Section III _ Pseudo Codes
- **Source File**: 1. Pseudo Codes Technical ebook.pdf
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

/ Code Snippet

```
find the output of the
following code.
#include<stdio.h>
int main(void)
{
int a=10, x;
X= a-- + +445
printf (“%d”,x);
return 0;
}
```

Options

- a. 19
- b. 20
- c. 22
- d. 23

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: Option B

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 136:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option B.

Visual Page Reference / Diagram



Problem 040 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(ID: `SEC03_EBOOK_Q138`)
- **Source Module**: `Section III _ Pseudo Codes`
- **Source File**: `1. Pseudo Codes Technical ebook.pdf`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** `Upper Level (Specialist Programmer / Digital Specialist Engineer)`
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

/ Code Snippet

```
find the output of the
following code.
#include<stdio.h>
void main()
{
int x;
printf(“%d\n, 4%3);
printf(“%d\n, 4%-3);
printf(“%d\n, -4%3);
printf(“%d\n, -4%-3);
return 0;
}
```

Options

- a. 1 1 -1 -1
- b. 1 -1 1 -1
- c. 1 -1 -1 1
- d. 1 -1 -1 -1

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Option A`

Step-by-Step Code Execution Trace & Proof

Execution Analysis & Formal Proof for Question 138:
1. Identify the base case and recurrence relation of the function.
2. Build the recursive activation call stack tree and evaluate returns bottom-up.
3. Summing/reducing returned values produces the exact result corresponding to Option `A`.

Visual Page Reference / Diagram



Problem 041 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** `Upper Level (Specialist Programmer / Digital Specialist Engineer)`
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

(ID: `SEC03_MOCK_Day04_Q10`)
- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 04 - Technical MCQ - Pseudo Codes Based on PYQ With Video Sol_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** `Upper Level (Specialist Programmer / Digital Specialist Engineer)`
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

/ Code Snippet

```
1.     What does the expression float a = 35 / 0 return?
```

Options

- 0
- Not a Number
- Infinity
- Run time exception

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Infinity`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 04 Video Solutions Mock. Full execution trace and operator evaluation confirms Option `Infinity`.

Visual Page Reference / Diagram



Problem 042 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** `Upper Level (Specialist Programmer / Digital Specialist Engineer)`
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

(ID: `SEC03_MOCK_Day04_Q12`)
- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 04 - Technical MCQ - Pseudo Codes Based on PYQ With Video Sol_QUESTIONS.json`
- **Topic Domain**: **Data Structures, Arrays & Pointers**

Question Statement 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** `Upper Level (Specialist Programmer / Digital Specialist Engineer)`
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

/ Code Snippet

```
1.    Construct post order Transversal for the constructed tree with the given In-Order and Preorder Travers
In-Order: 9 7 10 6 8
Pre-Order: 6 7 9 10 8
```

Options

- 9 10 7 8 6
- 9 7 6 10 8
- 8 9 10 7 6
- 9 8 10 7 6

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `9 10 7 8 6`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 04 Video Solutions Mock. Full execution trace and operator evaluation confirms Option `9 10 7 8 6`.

Visual Page Reference / Diagram



Problem 043 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** `Upper Level (Specialist Programmer / Digital Specialist Engineer)`
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

(ID: `SEC03_MOCK_Day04_Q15`)
- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 04 - Technical MCQ - Pseudo Codes Based on PYQ With Video Sol_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** `Upper Level (Specialist Programmer / Digital Specialist Engineer)`
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

/ Code Snippet

```
1.     What will be the output of this program?
#include
int main() {
int i = 5;
printf(“%d”, i = ++i == 6);
return 0;
}
```

Options

- 2
- 6
- 4
- 1

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `1`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 04 Video Solutions Mock. Full execution trace and operator evaluation confirms Option `1`.

Visual Page Reference / Diagram



Problem 044 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** `Upper Level (Specialist Programmer / Digital Specialist Engineer)`
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

(ID: `SEC03_MOCK_Day04_Q17`)
- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 04 - Technical MCQ - Pseudo Codes Based on PYQ With Video Sol_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** `Upper Level (Specialist Programmer / Digital Specialist Engineer)`
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

/ Code Snippet

```
1.     What will be the output of this program?
#include
enum flg{a, b, c};
enum glf{c, e, f};
main()
{
enum flg h;
h = b;
printf(“%d”, h);
return 0;
}
```

Options

- 1
- error: redeclaration of an enumerator
- h
- 3

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `error: redeclaration of an enumerator`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 04 Video Solutions Mock. Full execution trace and operator evaluation confirms Option `error: redeclaration of an enumerator`.

Visual Page Reference / Diagram



Problem 045 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** `Upper Level (Specialist Programmer / Digital Specialist Engineer)`
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

(ID: `SEC03_MOCK_Day04_Q21`)
- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 04 - Technical MCQ - Pseudo Codes Based on PYQ With Video Sol_QUESTIONS.json`
- **Topic Domain**: **Data Structures, Arrays & Pointers**

Question Statement 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** `Upper Level (Specialist Programmer / Digital Specialist Engineer)`
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

/ Code Snippet

```
1.     If the sequence of operations are performed on a stack, the sequence of pop will be?
Push(1), Push(2), Pop(), Push(1), Push(2), Pop(), Pop(), Pop(), Push(2), Pop().
```

Options

- 2 2 1 1 2
- 2 2 1 2 2
- 2 1 2 2 1
- 2 1 2 2 2

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `2 2 1 1 2`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 04 Video Solutions Mock. Full execution trace and operator evaluation confirms Option `2 2 1 1 2`.

Visual Page Reference / Diagram



Problem 046 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** `Upper Level (Specialist Programmer / Digital Specialist Engineer)`
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

(ID: `SEC03_MOCK_Day04_Q24`)
- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 04 - Technical MCQ - Pseudo Codes Based on PYQ With Video Sol_QUESTIONS.json`
- **Topic Domain**: **Data Structures, Arrays & Pointers**

Question Statement ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

/ Code Snippet

```
1.    Which of the following sequence can be obtained in the output (in same order) using a stack assuming
```

Options

- a) 3,4,5,1,2
- a) 3,4,5,2,1
- a) 1,5,2,3,4
- a) 5,4,3,1,2

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `a)  3,4,5,2,1`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 04 Video Solutions Mock. Full execution trace and operator evaluation confirms Option `a)  3,4,5,2,1`.

Visual Page Reference / Diagram



Problem 047 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(ID: `SEC03_MOCK_Day04_Q26`)
- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 04 - Technical MCQ - Pseudo Codes Based on PYQ With Video Sol_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

/ Code Snippet

```
1.     The value of j at the end of the execution of the following pseudo code is ?
Integer incr(Integer i)
Static Integer count = 0
count = count +i
return count
Main()
Integer i,j
for(start i from 0 to 4)
j= incr(i)
end for
```

Options

- 10
- 4

- 6
- 7

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `10`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 04 Video Solutions Mock. Full execution trace and operator evaluation confirms Option `10`.

Visual Page Reference / Diagram



Problem 048 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(ID: `SEC03_MOCK_Day04_Q28`)
- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 04 - Technical MCQ - Pseudo Codes Based on PYQ With Video Sol_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

/ Code Snippet

```
1.     The value returned by the following function for f(1) is?
int f(int n)
{
static int i =1;
if (n>=5) return n;
n=n + i ;
i++;
return f(n);
}
```

Options

- 5
- 6
- 7
- 8

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `7`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 04 Video Solutions Mock. Full execution trace and operator evaluation confirms Option `7`.

Visual Page Reference / Diagram



Problem 049 `◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]`

> 🎯 **Exam Priority:** `◯ [🕮 PRACTICE VARIANT / REDUNDANT]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(ID: `SEC03_MOCK_Day06_Q03`)
- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 06 - Pseudo Code_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement `◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]`

> 🎯 **Exam Priority:** `◯ [🕮 PRACTICE VARIANT / REDUNDANT]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

/ Code Snippet

```
Find the output of the following code. #include int main(void) { int a = 10, x; x = a-- + +445; printf ("%d'
```

Options

- 19
- 435

- 455
- 23

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `455`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 06 Full Length Mock I. Full execution trace and operator evaluation confirms Option `455`.

Visual Page Reference / Diagram



Problem 050 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(ID: `SEC03_MOCK_Day07_Q01`)
- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 07 - Section III _ Pseudo Codes Based on PYQ_QUESTIONS.json`
- **Topic Domain**: **Data Structures, Arrays & Pointers**

Question Statement ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

/ Code Snippet

```
What is the output of the following C code? #include int main() { int num = 987; int rem; while(num != 0) {
```

Options

- 0
- 1
- 10
- 5

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `1`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 07 Full Length Mock II. Full execution trace and operator evaluation confirms Option `1`.

Visual Page Reference / Diagram



Problem 051 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(ID: `SEC03_MOCK_Day07_Q02`)
- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 07 - Section III _ Pseudo Codes Based on PYQ_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

/ Code Snippet

```
What is the output of the following C code? #include int main() { float x = 0.0; long int y = 10; printf("%
```

Options

- 3
- 1
- 5
- 2

Correct Answer

> [!IMPORTANT]

> **Verified Answer Key**: `1`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 07 Full Length Mock II. Full execution trace and operator evaluation confirms Option `1`.

Visual Page Reference / Diagram



Problem 052 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(ID: `SEC03_MOCK_Day07_Q07`)
- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 07 - Section III _ Pseudo Codes Based on PYQ_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

/ Code Snippet

```
int main() { int x, y = 5, z = 5; x = y == z; Print(x) return 0; }
```

Options

- 0
- 1
- 5
- Compilation Error

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `1`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 07 Full Length Mock II. Full execution trace and operator evaluation confirms Option `1`.

Visual Page Reference / Diagram



Problem 053 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(ID: `SEC03_MOCK_Day08_Q06`)
- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 08 - Technical MCQ - Pseudo Codes Based on PYQ 01_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

/ Code Snippet

```
Q6. What is the output for p=15 and q=4. Integer solve (Integer p, Integer q) Integer value while(q) value =
```

Options

- 3
- 1
- 2
- 0

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `1`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 08 Full Length Mock III. Full execution trace and operator evaluation confirms Option `1`.

Visual Page Reference / Diagram



Problem 054 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(ID: SEC03_MOCK_Day08_Q07)
- **Source Module**: Infosys Mock Test Series
- **Source File**: Day 08 - Technical MCQ - Pseudo Codes Based on PYQ 01_QUESTIONS.json
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

/ Code Snippet

```
Q7.what is the output for number=4 int something(int number) { if(number <= 0) return 1; else return number
```

Options

- 12
- 24
- 1
- 20

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: 24

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 08 Full Length Mock III. Full execution trace and operator evaluation confirms Option 24.

Visual Page Reference / Diagram



Problem 055 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(ID: `SEC03_MOCK_Day08_Q09`)
- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 08 - Technical MCQ - Pseudo Codes Based on PYQ 01_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

/ Code Snippet

```
Q9. What is the output for x = 10, y = 5 Integer solve(Integer x, Integer y) if(y > 0) if(x > 0) return x +
```

Options

- undefined

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `undefined`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 08 Full Length Mock III. Full execution trace and operator evaluation confirms Option `undefined`.

Visual Page Reference / Diagram



Problem 056 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`

> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(ID: `SEC03_MOCK_Day09_Q04`)
- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 09 - Technical MCQ - Pseudo Codes Based on PYQ 02_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

/ Code Snippet

```
What is the output if fun(5) is called? Function fun(n) If n == 0 Return 0 Else Print(n) Return fun(n + 1) E
```

Options

- 1 2 3 4 5
- 5 4 3 2 1
- Stack overflow
- -1

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `Stack overflow`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 09 Full Length Mock IV. Full execution trace and operator evaluation confirms Option `Stack overflow` .

Visual Page Reference / Diagram



Problem 057 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

(ID: `SEC03_MOCK_Day09_Q05`)
- **Source Module**: `Infosys Mock Test Series`
- **Source File**: `Day 09 - Technical MCQ - Pseudo Codes Based on PYQ 02_QUESTIONS.json`
- **Topic Domain**: **Recursion & Function Call Stacks**

Question Statement ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

/ Code Snippet

```
int main() { int x, y = 5, z = 5; x = y == z; Print(x); return 0; }
```

Options

- 0
- 1
- 5
- Compilation Error

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: `1`

Step-by-Step Code Execution Trace & Proof

Official Video Solution from Day 09 Full Length Mock IV. Full execution trace and operator evaluation confirms Option `1`.

Visual Page Reference / Diagram


